relying on any such reference. It was as if someone gave the location for a party as 22 miles north without specifying the reference location, without specifying north of *what.* Most physicists, including Maxwell, attempted to explain the speed his equations gave in the following way: Familiar waves such as ocean waves or sound waves are carried by a substance, a medium. Ocean waves are carried by water. Sound waves are carried by air. And the speeds of these waves are specified *with respect to the medium.* When we talk about the speed of sound at room temperature being 767 miles per hour (also known as Mach 1, after the same Ernst Mach encountered earlier), we mean that sound waves travel through otherwise still air at this speed. Naturally, then, physicists surmised that light waves—electromagnetic waves—must also travel through some particular medium, one that had never been seen or detected but that must exist. To give this unseen light-carrying stuff due respect, it was given a name: the *luminiferous aether,* or the *aether* for short, the latter being an ancient term that Aristotle used to describe the magical catchall substance of which heavenly bodies were imagined to be made. And, to square this proposal with Maxwell's results, it was suggested that his equations implicitly took the perspective of someone at rest with respect to the aether. The 670 million miles per hour his equations came up with, then, was the speed of light relative to the stationary aether.

As you can see, there is a striking similarity between the luminiferous aether and Newton's absolute space. They both originated in attempts to provide a reference for defining motion; accelerated motion led to absolute space, light's motion led to the luminiferous aether. In fact, many physicists viewed the aether as a down-to-earth stand-in for the divine spirit that Henry More, Newton, and others had envisioned permeating absolute space. (Newton and others in his age had even used the term "aether" in their descriptions of absolute space.) But what actually *is* the aether? What is it made of? Where did it come from? Does it exist everywhere?

These questions about the aether are the same ones that for centuries had been asked about absolute space. But whereas the full Machian test for absolute space involved spinning around in a completely empty universe, physicists were able to propose doable experiments to determine whether the aether really existed. For example, if you swim